MIL-A-52873(ME)
17 July 1975

MILITARY SPECIFICATION

ALARM SET, ANTI-INTRUSION

(PSID) AN/GSQ-151

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a hand-emplaced, anti-intrusion seismic detection set.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-W-27076	- Workmanship Standards for Ground Electronic and Associated Equipment.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.

FSC 6350

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MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Enviromnental Test Methods.

DRAWINGS

ME

TA13219E9746	- Alarm Set, Anti-Intrusion (PSID) AN/GSQ-151.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The Alarm Set, Anti-Intrusion, AN/GSQ-151 (hereinafter called "PSID") shall be in accordance with Top Assembly TA13219E9746 and as specified herein. The PSID is an alarm system using seismic detection techniques, and consisting of four individual detectors which transmit indications, distinguishable from one another, of intrusion by personnel or vehicles by RF link to a receiver which is monitored by a headset. A PSID set consists

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of the above items and a carrying bag for the receiver and one for the complete system.

3.1.1 Drawings. All drawings forming a part of this specification are end product drawings and no deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more PSID's for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce PSID's that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier object to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more PSID's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Nonfungus nutrient materials shall be used where applicable. PSID's fabricated of materials other than those specified herein and any design changes required due to the use of such materials shall be subject to the approval of the contracting officer.

3.5 Environmental conditions. The PSID shall operate as specified herein without damage (see 6.4.1) when subjected to shock, vibration, temperature, humidity, and altitude extremes; salt fog; water immersion; and fungus growth as may be encountered in normal field usage when tested as specified herein.

3.5.1 Operating. The PSID set shall operate as specified herein without malfunction or damage while exposed to the following conditions:

(a) Temperature shocks within the range of +140° F to -22° F.
(b) Altitudes from sea level to 14,000 feet.
(c) Ambient temperature from +140° F with solar radiation of 360 Btu per square foot per hour, with not more than a 6 knot wind blowing, to -22° F without solar radiation, at all relative humidities possible at these temperatures.

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3.5.2 Storage. The PSID set shall operate as specified herein without malfunction or damage after exposure to the following conditions:

(a) Temperatures from -65° F to +155° F.
(b) Humidity up to 100 percent.
(c) Immersion to 3 feet when tested as specified in 4.6.5.4.
(d) Altitudes to 40,000 feet.
(e) Mechanical shock when tested as specified in 4.6.5.2.
(f) Vibration when tested as specified in 4.6.5.1.
(g) Fungus when tested as specified in 4.6.5.6.
(h) Salt fog when tested as specified in 4.6.5.5.

3.6 Technical requirements.

3.6.1 Electrical requirements (common to detector and receiver) - power supply. The detector and receiver shall operate from two 9-volt batteries, Type BA 1090/U, connected in parallel.

3.6.2 Electrical requirements (detector). Each detector shall consist of a geophone, amplifier and level detector, coding system, and transmitter.

3.6.2.1 Activation sequence. An activation sequence shall be initiated when the geophone is physically stimulated resulting in the generation of an electrical signal that is amplified and processed by the seismic processor. The seismic processor shall activate the coding system which shall cause the transmitter to turn on and emit a burst of RF energy modulated with audio frequency tone. The number of bursts and the audio frequency tone shall be controlled by the coding system.

3.6.3 Seismic system. The seismic system shall consist of a geophone, seismic bandpass amplifier, and threshold level detector.

3.6.3.1 Geophone. The geophone shall be resonant at 14 Hz and, with associated circuitry, detect seismic signals and activate a very high frequency (VHF) transmitter.

3.6.3.2 Seismic bandpass amplifier. The amplifier shall have a passband centered at a frequency of 20 $\pm$ 2 Hz. The 6 dB bandwidth shall be 8 $\pm$ 2 Hz to 40 $\pm$ 5 Hz. The high frequency response shall be down 15 dB at 75 $\pm$ 15 Hz.

3.6.3.3 Power gain switch. A six-position combination power switch and stepped gain control shall be provided for adjustment of amplifier sensitivity. The extreme counterclockwise position shall be the "OFF" position, and the

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maximum gain shall be obtained at the extreme clockwise position. Four steps of attenuation in 6 dB increments from the maximum gain position shall be provided.

3.6.3.4 Threshold level detector. The threshold level detector shall activate the coding system when a signal from the bandpass amplifier exceeds the detector's threshold. A sinusoidal signal with a frequency of 20 ± 2 Hz and a peak-to-peak amplitude of 72 microvolts plus 24 microvolts minus 48 microvolts applied to the amplifier input terminal shall result in the threshold being exceeded and the activation of the coding system. The gain selection control shall be in Position 5.

3.6.4 Coding system.

3.6.4.1 Coding pulse characteristics. Each of the four detector units shall have a distinct pulse code. Characteristics of the pulse codes shall be as follows over the operating temperature range:

(a) Detector Unit No. 1: One pulse at 500 Hz ± 50 Hz (square wave).
(b) Detector Unit No. 2: Two pulses at 1 kHz ± 100 Hz (square wave).
(c) Detector Unit No. 3: Three pulses at 250 Hz ± 25 Hz (square wave).
(d) Detector Unit No. 4: Four pulses at 2 kHz ± 200 Hz (square wave).

3.6.4.2 Message format. The transmitted message shall consist of pulses of 70 ± 10 milliseconds duration, with a 50 ± 10 millisecond spacing between individual pulses.

3.6.4.3 Recovery time. The coding system shall not accept additional activation signals for 100 milliseconds after transmission of a coded output. The maximum recovery time shall not exceed 1.8 seconds.

3.6.5 Transmitter.

3.6.5.1 Transmitter frequency. The RF output frequency shall be one of those shown in Table I, as specified (see 6.2). Total frequency error due to frequency setting tolerance of the transmitter and the frequency variation due to change in temperatures over the full operating range shall not exceed a total of ± 0.008 percent.

Table I. RF Frequencies (MHz)

126.600	133.500	148.71
127.400	134.500	149.73
129.000	148.02	150.57
129.900		

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3.6.5.2 Operating voltage range. The detector shall perform as specified herein with power supply voltages from 7.5 to 9.8 volts dc.

3.6.5.3 RF power output. Pulse output power shall be not less than 100 mw nor more than 450 mw as measured into a 50-ohm load.

3.6.5.4 Current drain. The detector under standby conditions shall draw not more than 1.5 milliamperes from its battery power supply.

3.6.6 Receiver.

3.6.6.1 Receiver frequency. The receiver frequency shall be as specified (see 6.2) from Table I.

3.6.6.2 Bandwidth. The IF bandwidth shall be a minimum of 36 kHz as measured at the 6 dB response points, and a maximum of 450 kHz as measured at the 60 dB response points.

3.6.6.3 Local oscillator frequency stability. The frequency setting tolerance of the local oscillator and the frequency variation with temperature shall not exceed a total of ± 0.008 percent over the operating temperature range.

3.6.6.4 Threshold sensitivity. The receiver response, with 100 percent modulated AM signal of the assigned receiver frequency with a modulation frequency of 400 Hz and an absolute signal level of -100 dbm applied to the receiver input terminals, shall be an output signal measured at the earphone connection into a 60-ohm load of at least 0.031 volt rms. The receiver shall exhibit a signal-to-noise ratio at this point of at least 12 dB.

3.6.6.5 Dynamic range. The receiver shall be capable of operating normally with signal inputs between -100 dbm and -40 dbm into the receiver input terminals. At -40 dbm the receiver output as measured at the earphone connection into a 60-ohm load shall not exceed 0.5 volt rms.

3.6.6.6 Image rejection. The image rejection of the receiver shall be at least 60 dB.

3.6.6.7 Spurious response rejection. Response to any signal outside the passband of the receiver shall be at least 60 dB below signals occurring in the desired passband.

3.6.6.8 Adjacent channel rejection. RF signals 600 kHz or more from the assigned channel frequency shall be at least 60 dB below the output obtained from a signal at the assigned channel frequency.

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3.6.6.9 Overload. The receiver shall be capable of sustaining an overload of 1.0 volt rms applied directly to the antenna terminals of the receiver at both 60 Hz and at channel frequency. The overload shall not damage the receiver, nor in any way alter its operation.

3.6.6.10 Audio frequency response. The variation of the audio output at the earphone connector into a 60-ohm resistive load shall be not more than ± 3 dB for modulation frequencies from 250 Hz to 3 kHz.

3.6.6.11 Recovery time. The receiver shall be fully operational at normal sensitivity within 50 milliseconds after application of a 70 millisecond pulse of RF energy at the assigned channel frequency with a pulse signal strength of -40 dbm as measured at the receiver input terminals.

3.6.6.12 Operating voltage range. The receiver shall perform as specified herein with power supply voltages from 7.0 to 9.8 Vdc.

3.6.6.13 Current drain. The receiver under standby conditions shall draw not more than 10 milliamperes from its battery power supply.

3.6.6.14 Receiver operation. The receiver shall be capable of receiving the coded messages specified in 3.6.4.

3.7 Electromagnetic interference. The PSID electromagnetic interference emission and susceptibility characteristics shall conform to MIL-STD-461, Class IA for the receiver, and Class ID for the detector.

3.8 Reliability. The specified Mean-Time-Between-Failure (MTBF) of the receiver shall be 6800 hours, and of the detector shall be 6800 hours, when tested as specified in 4.6.5.8.

3.9 Burn-in. The PSID shall be capable of operation without failure for 48 consecutive hours while activated 100 times per day.

3.10 Workmanship. Workmanship shall be not less than that specified in MIL-W-27076. All parts, components and assemblies of the PSID, including castings, forgings, molded parts, stamping, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, sprues, scale, flux, weld spatter, and other harmful extraneous materials. External surfaces shall be free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of

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all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction PSID shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction PSID shall be subjected to the tests marked "X" in Column 1 of Table II. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production PSID's will be selected at random by the Government from the PSID's being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the

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Government. Acceptance of an initial production PSID shall not exclude the remaining PSID's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production PSID to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production PSID and shall be cause for refusal by the Government to continue acceptance of production PSID's until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on PSID's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all PSID's accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.5.2 Examination. Each PSID set shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.3 Tests.

4.5.3.1 Individual. Each PSID set shall be subjected to the tests marked "X" in Column 3 of Table II. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table II. AQL shall be 1.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The PSID set shall be examined as specified herein for the following defects:

101. Protective treatment not as specified.
102. Metals not as specified.

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103. Printed wiring assemblies and boards not as specified.
104. Fungus and moisture resistance not as specified.
105. Aluminum castings not as specified.
106. Component parts not as specified.
107. Construction not as specified.
108. Antenna not as specified.
109. Power-gain switch not as specified (detector).
110. Dimensions not as specified.
111. Treatment and painting not as specified.
112. Identification and instruction markings not as specified.
113. Electromagnetic interference suppression components and network not identical to the approved model.
114. Workmanship not as specified.

4.6.2 Test schedule.

Table II. Test Schedule

Preproduction	Initial Production and Inspection Comparison	Quality Conformance Individual	Quality Conformance Samples	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
				Detector		
X	X	X	X	Activation Sequence	4.6.3.11	3.6.2.1
X	-	X	-	Seismic Bandpass Amplifier	4.6.3.2	3.6.3.2
X	-	X	-	Power Gain Switch	4.6.3.3	3.6.3.3
X	-	X	X	Threshold Level Detector	4.6.3.4	3.6.3.4
X	X	X	X	Code Pulse Characteristics	4.6.3.5	3.6.4.1
X	X	X	X	Message Format	4.6.3.6	3.6.4.2
X	X	X	-	Recovery Time	4.6.3.7	3.6.4.3
X	X	X	X	Transmitter Frequency	4.6.3.8	3.6.5.1
X	X	X	X	RF Power Output	4.6.3.9	3.6.5.3
X	X	X	X	Current Drain	4.6.3.10	3.6.5.4

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Table II. Test Schedule (Cont'd)

1	2	3	4	5	6	7
				Reciver		
X	X	X	X	Receiver Frequency	4.6.4.1	3.6.6.1
X	-	X	-	Bandwidth	4.6.4.3	3.6.6.2
X	-	-	-	Local Oscillator Frequency Stability	4.6.4.2	3.6.6.3
X	X	X	X	Threshold Sensitivity	4.6.4.1	3.6.6.4
X	X	X	-	Dynamic Range	4.6.4.4	3.6.6.5
X	X	-	X	Image Rejection	4.6.4.5	3.6.6.6
X	X	-	-	Spurious Response Rejection	4.6.4.6	3.6.6.7
X	X	-	X	Adjacent Channel Rejection	4.6.4.7	3.6.6.8
X	X	X	-	Overload	4.6.4.8	3.6.6.9
X	X	X	X	Audio Frequence Response	4.6.4.9	3.6.6.10
X	X	X	-	Recovery Time	4.6.4.10	3.6.6.11
X	X	X	-	Receiver Operation	4.6.4.12	3.6.6.14
X	X	X	X	Current Drain	4.6.4.11	3.6.6.13
X	=	=	-	EMI	4.6.5.7	3.7
X	X	-	-	Reliability	4.6.5.8	3.8
X	X	-	X	Vibration	4.6.5.1	3.5
X	X	-	-	Mechanical Shock	4.6.5.2	3.5
X	X	-	X	High Temperature and Humidity Storage	4.6.6.1	3.5
X	X	-	-	Temperature Operation and Thermal Shock	4.6.6.2	3.5
X	X	-	-	Altitude - Temperature	4.6.5.3	3.5
X	X	-	X	Immersion	4.6.5.4	3.5
X	X	-	-	Salt Fog	4.6.5.5	3.5
X	X	-	-	Fungus Growth	4.6.5.6	3.5

4.6.3 Tests.

4.6.3.1 Detector. Unless otherwise specified herein, the following tests shall be performed at 75° ± 5° F, with a power supply voltage of 7.5 and 9.8 volts dc, with the antenna removed, and the detector output terminated in a 50-ohm load. Tests specified in 4.6.3.2, 4.6.3.3, 4.6.3.4, 4.6.3.6, and 4.6.3.7 may be performed in process prior to sealing the units. For quality conformance sample testing, the test specified in 4.6.3.4 may be performed by comparison with the preproduction unit under the same seismic input conditions.

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4.6.3.2 Seismic bandpass amplifier. Measure the amplifier bandwidth with the geophone disconnected. Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.6.3.3 Power gain switch. Measure the change in gain for each sensitivity switch position. Nonconformance to 3.6.3.3 shall constitute failure of this test.

4.6.3.4 Threshold level detector. Place the sensitivity switch in Position 5. Supply a 20 ± 2 Hz sinusoidal signal with peak-to-peak amplitude of 72 microvolts (plus 24 microvolts minus 48 microvolts) to geophone input terminals of the amplifier (geophone disconnected). The coding system should turn "on" the transmitter and transmit a message. Nonconformance to 3.6.3.4 shall constitute failure of this test.

4.6.3.5 Code pulse characteristics. Activate the detector (see 3.6.2.1). Measure the number of pulses and frequency of pulses transmitted by the detector. The number of pulses shall be equal to the number of raised dots on the detector case and the frequency shall be in accordance with 3.6.4.1. Nonconformance to 3.6.4.1 shall constitute failure of this test.

4.6.3.6 Message format. Activate the detector (see 3.6.2.1). Measure the spacing between pulses in the message. Nonconformance to 3.6.4.2 shall constitute failure of this test.

4.6.3.7 Recovery time. Activate the detector (see 3.6.2.1). Measure the time from the trailing edge of the last pulse to the beginning of the next series of pulses. Nonconformance to 3.6.4.3 shall constitute failure of this test.

4.6.3.8 Transmitter frequency. Activate the transmitter and measure the output frequency. The frequency shall be on the same channel as indicated on the detector identification plate. Nonconformance to 3.6.5.1, or the detector not on the frequency marked on the identification plate shall constitute failure of this test.

4.6.3.9 RF power output. Activate the detector (see 3.6.2.1). Measure the power output into a 50-ohm load. Nonconformance to 3.6.5.3 shall constitute failure of this test.

4.6.3.10 Current drain. Measure the current of the detector under standby conditions. The current drain shall be measured at a power supply of 9.8 Vdc. Nonconformance to 3.6.5.4 shall constitute failure of this test.

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4.6.3.11 Activation sequence. Remove the 50-ohm load and install the antenna and supply power with two 9-volt batteries. Place the gain control in Position 5. Isolate the geophone from the ambient seismic disturbances. Supply a controlled stimulus to the geophone and monitor the detector RF output. The detector shall activate and transmit a message. Repeat for each gain position. Nonconformance to 3.6.2.1 shall constitute failure of this test.

4.6.4 Receiver. Unless otherwise specified herein, the following tests shall be performed at 75° ± 5° F with power supply voltage of 7.5 and 9.8 volts dc, with the antenna removed, and the output terminated in a 60-ohm load. A standard test signal shall be defined as a signal with 100 percent amplitude modulation (AM) of 400 Hz at the frequency specified on the receiver identification plate; the signal input level shall be -100 dbm. The test specified in 4.6.4.3 may be performed in process prior to sealing the units.

4.6.4.1 Receiver frequency and threshold sensitivity. Connect the standard test signal to the receiver antenna terminals. Measure the receiver output. Remove the modulation and measure the receiver output. The receiver output shall be 12 dB lower. Nonconformance to 3.6.6.1 and 3.6.6.4 shall constitute failure of this test.

4.6.4.2 Local oscillator stability. Measure the frequency of the local oscillator. Nonconformance to 3.6.6.3 shall constitute failure of this test.

4.6.4.3 Bandwidth. Measure the IF bandwidth. Nonconformance to 3.6.6.2 shall constitute failure of this test.

4.6.4.4 Dynamic range. Connect the standard test signal to receiver antenna terminal. Measure the receiver output. Increase the input signal level to -40 dbm. Measure the receiver output. Nonconformance to 3.6.6.5 shall constitute failure of this test.

4.6.4.5 Image rejection. Supply the standard test signal to the antenna terminal of the receiver. Reduce the signal level to the minimum receiver output voltage. Record the receiver output voltage. Adjust the input signal to the image frequency. Reduce the input signal level to the minimum receiver output voltage. Nonconformance to 3.6.6.6 shall constitute failure of this test.

4.6.4.6 Spurious response rejection. Supply the standard test signal to receiver antenna terminals. Measure the receiver output voltage Increase the signal level by 60 dB. Vary the input frequency above and below

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the channel frequency and record the receiver output voltage. Nonconformance to 3.6.6.7 shall constitute failure of this test.

4.6.4.7 Adjacent channel rejection. Supply the standard test signal to the receiver antenna terminals. Measure the receiver output voltage. Increase the signal level by 60 dB. Change the frequency to 600 kHz above the channel frequency. Record the receiver output voltage. Repeat the above test for the frequency 600 kHz below the channel frequency and record the receiver output voltage. Nonconformance to 3.6.6.8 shall constitute failure of this test.

4.6.4.8 Overload. Supply 1.0 volt rms at 60 Hz to the antenna terminals for 10 seconds. Remove the signal and conduct the test specified in 4.6.4.1. Supply 1.0 volt rms at the channel frequency to the antenna terminals. Remove the signal and conduct the test specified in 4.6.4.1. Nonconformance to 3.6.6.9 shall constitute failure of this test.

4.6.4.9 Audio frequency response. Supply the standard test signal. Measure the receiver output voltage. Adjust the modulation frequency over the range from 250 Hz to 3 kHz and record the receiver output voltages. Nonconformance to 3.6.6.10 shall constitute failure of this test.

4.6.4.10 Recovery time. Supply the standard test signal for 70 milliseconds to the antenna terminals. Measure the time from the end of the pulse to the receiver output noise level (no signal input). Nonconformance to 3.6.6.11 shall constitute failure of this test.

4.6.4.11 Current drain. Measure the receiver current drain from the battery power supply at 9.8, 8.5, and 7.5 volts dc. Nonconformance to 3.6.6.13 shall constitute failure of this test.

4.6.4.12 Receiver coding operation. The receiver shall be tested with the antenna and headset connected to the receiver. Activate four detectors as specified in 4.6.3.11; observe the receiver receiving the coding specified in 3.6.4.1 transmitted by the detectors. Nonconformance to 3.6.6.14 shall constitute failure of this test.

4.6.5 Environmental. Upon completion of all environmental testing, the PSID sets shall be subjected to the tests marked "X" in Column 3 of Table II except for those tests performed in process prior to sealing the unit.

4.6.5.1 Vibration. The PSID set shall be subjected to an acceleration of 5 g over a frequency range from 30 to 2000 to 30 Hz swept logarithmically over a 20-minute period in three mutually perpendicular planes. After completion

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of the test, the PSID set shall be examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Evidence of any damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.6.5.2 Mechanical shock. The PSID set shall be tested in accordance with MIL-STD-810, Method 516, Procedure I, for sawtooth pulse, peak value 20 g, duration 11 ms. The headset shall be tested in accordance with MIL-STD-810, Method 516, Procedure I, for sawtooth pulse, peak value 20 g, duration 11 ms. After completion of the test, the PSID set shall be examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Evidence of any damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.6.5.3 Altitude - temperature.

4.6.5.3.1 Operation. The PSID set shall be placed within an altitude test chamber and the internal absolute pressure reduced to 17.5 inches of mercury (corresponding to an altitude of 14,000 feet above sea level) for 1 hour. The temperature shall be room temperature. Perform the tests specified in 4.6.3.11 and 4.6.4.12. Any evidence of damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.6.5.3.2 Storage. The PSID set shall be placed within an altitude test chamber, the temperature of which shall be stabilized at -65° F without benefit of solar radiation. The internal absolute pressure shall be reduced to 5.5 inches of mercury (corresponding to an altitude of 40,000 feet above sea level). This pressure shall be maintained for 4 hours. The pressure in the chamber shall then be increased at a rate of not less than 3.0 inches of mercury per minute until atmospheric pressure has been reached. Upon removal from the chamber the PSID set shall be examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Any evidence of damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.6.5.4 Immersion. The PSID set shall be subjected to the leakage test in accordance with MIL-STD-810, Method 512, Procedure I, except that the temperature difference between the water and the PSID set at the time of submersion shall be not less than 10° C. The water shall be cooler. The headset shall be tested separately. The water shall be made conductive by being made a saturated solution using sodium chloride. Bubbles which are the result of entrapped air on the exterior of component parts shall not be considered a leak. Within 30 minutes after completion of the test, the PSID shall be visually inspected for entrance of water, examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. The visual inspection shall include the battery compartment. Any evidence of damage (see 6.4.1) or leakage, or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

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4.6.5.5 Salt fog. The PSID set with all covers, seals, and protective devices removed as required for maintenance settings or adjustments shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509, Procedure I. After completion of the test, the PSID shall be inspected for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Any evidence of damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.6.5.6 Fungus growth. The PSID set shall be placed within a tropical testing chamber (see 6.4) and subjected to exposure under simulated cyclic tropical conditions prevailing therein for a period of 90 days. Environmental conditions within the chamber shall be as follows:

> The tropical testing chamber shall operate on a daily temperature humidity cycle typical of a tropical area. For an 8-hour period, the temperature shall be held at 85° F and the relative humidity at 70 percent. During the next 8 hours of the cycle, the temperature shall be gradually lowered to 75° F and the relative humidity then kept at 75° F and 95 percent for a 4-hour period, followed by a gradual change to 85° F and 70 percent during the next 4-hour period, which completes the cycle.

The chamber atmosphere shall be heavily charged with spores of various species of fungi and provide continuous inoculation of exposed items and materials to such an extent that artificial prior inoculation with fungus spores is unnecessary.

The following list is representative of the fungi present in the chamber:

Acremonium alternatum	Memoniella echinata
Ascotricha xylina	Mucor sp.
Aspergillus fluvus	Myrothecium verrucaria
Aspergillus niger	Nigrospora sphaerica
Aspergillus oryzae	Penicillium citrinum
Aspergillus versicolor	Penicillium luteum
Aspergillus terreus	Penicillium purpurogenum
Chaetomium bostrychodes	Penicillium stoloniferum
Chaetomium indicum	Pollularia sp.
Chaetomium globosum	Rhizopus nigricans
Coniothyrium sp.	Spicaria sp.
Curvularia sp.	Stachbotrys atra
Fusarium sp.	Trichoderma viride

Upon removal from the chamber after the 90-day exposure period, the PSID set shall be examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Any evidence of damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

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4.6.5.6.1 For initial production inspection. For initial production inspection, the fungus growth Test Method No. 508 of MIL-STD-810 shall be used in lieu of 4.6.5.6.

4.6.5.7 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.7. The supplier shall furnish the contracting officer with a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6.5.8 Reliability. When specified (see 6.2), using the respective MTBF's for the receiver and the detector specified in 3.8, the PSID set shall be tested as specified herein, with accept and reject criteria in accordance with MIL-STD-781, Test Plan VI. The time frame for conducting the test and the number of units to be tested shall be as specified (see 6.2). Items shall have been "burned-in" as specified in 3.9 and shall have passed the individual quality conformance inspection specified in 4.5.3.1. Testing shall consist of continuous operation of the PSID set with 100 activations of each detector each 24 hour period. The test shall be conducted under ambient environmental conditions. Testing shall continue until an accept or reject decision is reached. For the purpose of this test, a failure is defined as an event in which malfunction occurs resulting in the replacement of the malfunctioning material by a like item, the removal and repair of the malfunctioning item, or any repairs required because of faulty workmanship in manufacture. Any functional variation outside the limits specified herein or any impending malfunction detected which constitutes a safety hazard to operating personnel or would cause or may cause serious damage to the set if continued in operation is considered to be a failure. Normal replacement of batteries during test will not be considered a reliability-chargeable failure. Nonconformance to 3.8 shall constitute failure of this test.

4.6.6 Environmental.

4.6.6.1 High temperature and humidity storage. PSID sets shall be tested in accordance with MIL-STD-202, Method 106, except that the upper temperature limit shall be 155° F and the humidity test shall be 11 cycles for preproduction models and 3 cycles for production samples. Upon completion of Step "b" of the final cycle and after completion of the test, the PSID set shall be examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Any evidence of damage or inability to perform as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.6.6.2 Temperature operation and thermal shock. The PSID sets shall be placed in a temperature chamber, the temperature of which is 140° F with solar radiation, and allowed to stabilize for 120 minutes. Perform the tests

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specified in 4.6.3.11 and 4.6.4.12. Lower the temperature in the chamber to +100° F and allow to stabilize for 60 minutes without solar radiation. Perform the tests specified in 4.6.3.11 and 4.6.4.12. Lower the temperature in the chamber to +25° F and allow to stabilize for 60 minutes without solar radiation. Perform the tests specified in 4.6.3.11 and 4.6.4.12. Lower the temperature in the chamber to 0° F and allow to stabilize for 60 minutes without solar radiation. Perform the tests specified in 4.6.3.11 and 4.6.4.12. Lower the temperature to -22° F and allow to stabilize for 60 minutes without solar radiation. Perform the tests specified in 4.6.3.11 and 4.6.4.12. Remove the PSID from the chamber to room temperature (75° F ± 5° F). After 30 minutes after removal from the chamber, the PSID set shall be examined for damage and tested as specified in 4.6.3.11 and 4.6.4.12. Any evidence of damage or inability to perform after each test as specified in 4.6.3.11 or 4.6.4.12 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select PSID sets at any time during the contract production period and subject these PSID sets to the examination specified in 4.6.1 and the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production PSID. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison PSID to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison PSID and shall be cause for refusal by the Government to continue acceptance of production PSID's until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on PSID's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all PSID's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

115.	Materials, methods,and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
116.	Components not cushioned and secured within the box as specified for Level A.
117.	Box not sealed as specified for Level A.
118.	Gross weight exceeds 200 pounds for Level A or B.
119.	Packing not as specified for Level A or B.
120.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. The PSID set contained in the carrying case shall be preserved in accordance with MIL-P-116, Method IIb. The inner container shall conform to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 125. The contents shall be cushioned and secured within the box in a manner to prevent movement or damage. The outer container shall conform to PPP-B-636, W6c. Closure and sealing of each of the boxes shall be in accordance with the appendix to the applicable box specification, using Method V on the exterior box and Method I or II on the inner box.

5.1.2 Level C. The PSID set contained in the carrying case shall be packaged in a fiberboard box and cushioned and secured in a manner to afford protection against mechanical or physical damage from the supplier to the initial destination.

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5.2 Packing. Packing shall be Level A, B,or C, as specified (see 6.2).

5.2.1 Level A. The PSID sets preserved and packaged as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, Grade B, in quantities not to exceed 200 pounds. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. PSID sets preserved and packaged as specified in 5.1 shall be packed as specified in 5.2.1 for Level A, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. The PSID sets preserved and packaged as specified in 5.1 shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The PSID set is a multiple-unit, portable, expendable intrusion alarm system capable of rapid installation, designed to protect area perimeters against personnel and vehicular intrusion.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number,and date of this specification.
(b) Time frame required for submission of preproduction model and number of PSID's required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required, and the number of PSID sets to be furnished (see 3.3).
(e) Transmitter and receiver frequencies required (see 3.6.5.1).
(f) When a reliability test is required (see 4.6.5.8).
(g) Time frame required for conducting reliability test, and the number of PSID's required (see 4.6.5.8).
(h) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

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6.3 Preproduction model. Any changes or deviations of preproduction PSID sets from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction PSID sets will not relieve the supplier of his obligation to furnish PSID sets conforming to this specification.

6.4 Definition.

6.4.1 Damage. Damage includes malfunction, deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the PSID with the provisions of this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, wiring diagrams and supplier's maintenance and operation manual to be furnished with each PSID.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of PSID's.

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DOCUMENT IDENTIFIER AND TITLE: MIL-A-52873(ME) Alarm Set, Anti-Intrusion (PSID) AN/GSQ-151

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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